Figure 2: Shades of blue

The image in Figure 2 is the output from the above example.

R has a number of built-in numeric functions. A few examples (square root, absolute value, floor, ceiling, truncate, cosine, exponent) with their respective outputs are shown below:

```
> sqrt(2)
[1] 1.414214

> abs(-3)
[1] 3

> floor(5.67)
[1] 5

> ceiling(5.67)
[1] 6

> trunc(4.32)
[1] 4

> cos(0)
[1] 1

> exp(1)
[1] 2.718282
```

There are also predefined functions (to upper case, to lower case, grep, string split) which operate on characters that you can use as follows:

```
> toupper('project')
[1] "PROJECT"

> tolower('LOWER')
[1] "lower"

> grep('l', 'lower')
[1] 1

> grep('l', 'upper')
integer(0)

> strsplit("0,Item,Quantity,GST", ",")
[[1]]
[1] "0"        "Item"      "Quantity" "GST"
```

Since R is designed for statistical computing, there are also built-in statistical functions (sum, minimum, maximum, range, mean, median) available. A few examples are shown below:

```
> sum(1, 2, 3)
[1] 6

> min(1, 2, 3)
[1] 1

> max(1, 2, 3)
[1] 3

> range(1, 2, 3)
[1] 1 3

> x <- c(1, 2, 3)

> mean(x)
[1] 2

> median(x)
[1] 2
```

You can load a library into the R runtime environment using the *library* function. We will now import the *Lattice* library in R, which is useful for visualising data:

```
> library(lattice)
>
```

There also exists a 'citation' function that gives you information on how to cite R or its packages when mentioning it in publications. The output for the same is shown below for reference:

```
> citation()

To cite R in publications use:

  R Core Team (2021). R: A language and environment for
statistical
  computing. R Foundation for Statistical Computing, Vienna,
Austria.
  URL https://www.R-project.org/.
```